

2026/1/28

On anomalies and
fermionic unitary operators

MO, Shimamura, Tachikawa, Yi (2025)



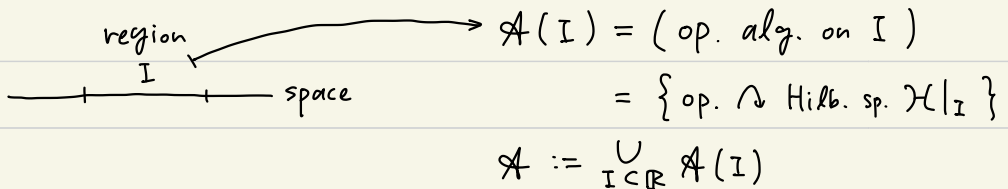
1. Anomalies of fin. grp. sym. of 1+1d QFT

obstruction to gauging, orbifold

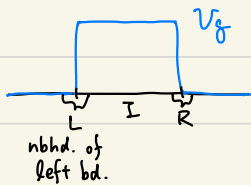
1.1. Bosonic case

Hamiltonian formalism $\left[\begin{array}{l} \text{phys.: Else-Nayak (2014), }^{\text{lattice}} \text{Seifnashri (2023)} \\ \text{AQFT, local cfm. net} \\ \text{math.: Müger (2004)} \end{array} \right]$

• 1+1d QFT



• sym. action of $g \in G$ on I (G : fin. grp. sym.)



autom. $\rho_g : \mathcal{A} \rightarrow \mathcal{A}$

$$\begin{array}{ccc} \mathcal{A} & \rightarrow & \mathcal{A} \\ \cup & & \cup \\ 0 & \mapsto & V_g 0 (V_g)^{-1} \end{array}$$

• fusion op.

$$\rho_g \rho_h(0) = U_{g,h} \rho_{gh}(0) (U_{g,h})^{-1}$$

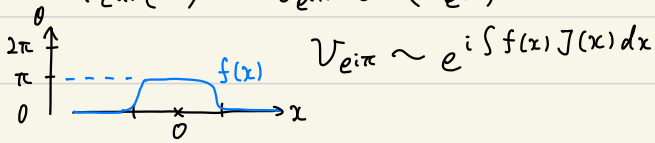
$$U_{g,h} = \underbrace{U_{g,h}^L}_{\mathcal{A}(L)} \cdot \underbrace{U_{g,h}^R}_{\mathcal{A}(R)} \quad \text{unitary "fusion op."}$$

$$V_g V_h = U_{g,h} V_{gh}.$$

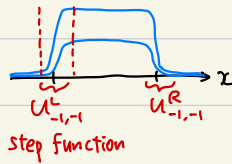
If $0 \in \mathcal{A}(I)$ s.t. $\begin{array}{l} I \cap L = \emptyset \\ I \cap R = \emptyset \end{array}$, then by locality, $\rho_g \rho_h(0) = \rho_{gh}(0)$.

e.g. $G = \mathbb{Z}_2 = \{e^{i0}, e^{i\pi}\} \subset U(1)$

$$\rho_{e^{i\pi}}(0) = V_{e^{i\pi}} 0 (V_{e^{i\pi}})^{-1}$$



$$\rho_{e^{i\pi}} \rho_{e^{i\pi}}(0) = U_{-1,-1}^L U_{-1,-1}^R 0 (U_{-1,-1}^R)^{-1} (U_{-1,-1}^L)^{-1}$$



anomaly

For any $O \in \mathcal{A}(L)$, $O \in \mathcal{A}(I)$, $I \cap L = \emptyset$, $I \cap R = \emptyset$ $I \not\subseteq \mathbb{R} \Rightarrow U_{h,k} \in \mathcal{P}_{hk}(O)$ $\nexists$ commute.

$$\begin{aligned} \rho_g(\rho_h \rho_k(O)) &= \rho_g(U_{h,k} \rho_{hk}(O) U_{h,k}^{-1}) \\ &= \underbrace{\rho_g(U_{h,k}) U_{g,hk} \rho_{ghk}(O) U_{g,hk}^{-1}}_{\rho_g(U_{h,k})^{-1} \rho_g(U_{h,k})} \rho_g(U_{h,k})^{-1} \end{aligned}$$

$$\begin{aligned} (\rho_g \rho_h)(\rho_k(O)) &= U_{g,h} \rho_{gh}(\rho_k(O)) U_{g,h}^{-1} \\ &= \underbrace{U_{g,h} U_{gh,k} \rho_{ghk}(O) U_{gh,k}^{-1}}_{\rho_{ghk}(O)} U_{g,h}^{-1} \end{aligned}$$

$\Downarrow \mathcal{A}(L) : \text{factor } [U_{g,h} U_{gh,k}]^{-1} \rho_g(U_{h,k}) U_{g,hk} \text{ commutes with } \forall O \in \mathcal{A}(L)$

$\exists \alpha : G \times G \times G \rightarrow V(1) \text{ s.t.}$

$$\rho_g(U_{hk}) U_{g,hk} = \alpha(g,h,k) U_{g,h} U_{gh,k} \quad \dots (\star)$$

ρ_k on BHS of $(\star) \rightsquigarrow \delta\alpha = 0$ (abusing notation! $\alpha = e^{2\pi i \alpha}$)

(pentagon identity) $\alpha(g,h,k) \alpha(lg,h,k)^{-1} \alpha(l,g,h,k) \alpha(l,g,hk)^{-1} \alpha(l,g,h) = 1$

redefinition $\tilde{U}_{g,h} := \tau(g,h) U_{g,h} \rightsquigarrow \tilde{\alpha} = \alpha + \delta\tau$

In this way, a cohomology class $[\alpha] \in H^3(G; V(1))$

"anomaly"

is associated to the QFT.

obstruction to gauging $\alpha \stackrel{?}{=} 0$

$$\begin{array}{c} U_{g,hk} \\ \swarrow \quad \searrow \\ g \quad h \quad k \end{array} U_{hk} = \alpha(g,h,k) \begin{array}{c} U_{g,h} \quad U_{gh,k} \\ \swarrow \quad \searrow \\ g \quad h \quad k \end{array}$$

If $[\alpha] = 0$, then G is anomaly-free.

1.2. Fermionic case

$1+1d$ $\left\{ \begin{array}{l} \text{bosonic} \\ \text{fermionic} \end{array} \right\}$ anomalies are classified by $\left\{ \begin{array}{l} H^3(G; \mathbb{Z}_2) \\ SH(G) \text{ "supercohomology"} \end{array} \right.$

$$SH(G) \stackrel{\text{as a set}}{=} H^3(G; \mathbb{Z}_2) \times H^2(G; \mathbb{Z}_2) \times H^1(G; \mathbb{Z}_2)$$

(e.g. $G = \mathbb{Z}_2$ $\mathbb{Z}_8$ $\left(\begin{array}{ccc} \mathbb{Z}_2 & \mathbb{Z}_2 & \mathbb{Z}_2 \end{array} \right)$)

bosonic part fermionic parity of fusion op. nontriv. when # real ferm. is odd (not considered in our study)

When $O^L \in \mathcal{A}(L)$,

$$\begin{aligned} P_g P_h(O^L) &= U_{g,h}^L U_{g,h}^R P_{gh}(O^L) (U_{g,h}^R)^{-1} (U_{g,h}^L)^{-1} \\ &= |U_{g,h}^R| |P_{gh}(O^L)| U_{g,h}^L P_{gh}(O^L) (U_{g,h}^L)^{-1} \\ &\quad \left| \begin{array}{l} P_g(\bullet) = \mathcal{V}_g \bullet \mathcal{V}_g^{-1} \quad \mathcal{V}_g : \text{bosonic} \left(\begin{array}{l} \text{shrink } I \rightarrow 0 \\ \mathcal{V}_g \rightarrow \text{id} \end{array} \right) \\ \Rightarrow |U_{g,h}^R| = |U_{g,h}^L|, \quad |P_{gh}(O^L)| = |O^L| \end{array} \right. \\ &= |U_{g,h}^L| |O^L| U_{g,h}^L P_{gh}(O^L) \end{aligned}$$

$$\mathcal{V}(g,h) := |U_{g,h}^L|$$

· Discussion similar to $(\star) \leadsto \delta \mathcal{V} = 0$.

· redefinition $\tilde{\mathcal{V}}_g = \Sigma_g^L \Sigma_g^R \mathcal{V}_g \leadsto \tilde{\mathcal{V}} = \mathcal{V} + \delta \xi \quad (\xi(g) := |\Sigma_g^L|)$

$$\therefore [\mathcal{V}] \in H^2(G; \mathbb{Z}_2)$$

($\delta \alpha = 0$ and $\alpha \sim \alpha + \delta \tau$ are modified by $\mathcal{V}$ and ξ .)

$$(\delta \alpha)^2 = \mathcal{V} \cdot \mathcal{V}$$

2. Values of α and V for $G \subset U(1)$

It is often the case that

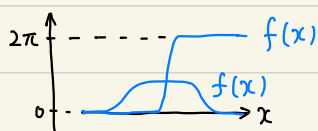
the cohomology class $[(\alpha, V)] \in SH(G)$ is known

but the values of cocycles α, V are not known.

→ cause problems in calculation of the partition functions of orbifolds (gauged theory)

In the case of $G \subset U(1)$, we succeeded in calculating α, V .

To do so, we introduce $U(f) := e^{i \int f(x) J(x) dx}$



$J(x)$: level- k $U(1)$ current

$$[J(x), J(y)] = k \frac{i}{2\pi} \partial_y \delta(x-y)$$

In particular, to calculate α, V , we need the commutator γ

$$U(f)U(g) = \underbrace{\gamma(f, g)}_{\substack{\uparrow \\ U(1)}} U(g)U(f).$$

because we need $P_g(U_{n,k}) = U_g U_{n,k} U_g^{-1}$ for $(\star\star)$.

Main difficulty ← compactified (because $f(-\infty) = f(\infty)$)

When $f: S^1 \rightarrow U(1)$ has non-zero winding number,

it is difficult to make $U(f)$ well-defined.

$f: \mathbb{R} \rightarrow U(1)$ だと $\int f dx = \infty$ だし

$f: S^1 \rightarrow U(1)$ とあっても S^1 を $[0, 2\pi]$ などで param. したときに必ず不連続な jump が発生

Strategy 1: Introduce patches to S' and make $u(f)$ well-def.

Strategy 2: Impose some requirements from physics on γ , and determine γ uniquely.

requirements

$$\mathcal{F} := \{ S' \rightarrow V(1) \text{ conti.} \}$$

$$\gamma : \mathcal{F} \times \mathcal{F} \rightarrow V(1) \text{ satisfies}$$

- bilinear and alternating $\gamma(f, f) = 0$.
- locality $\gamma(f, g) \in \{\pm 1\}$ if $\text{supp } f \cap \text{supp } g = \emptyset$.
- $\gamma(f_0, g_0) = \exp \left[2\pi i \frac{k}{2} \int_{S'} (f_0 g'_0 - g_0 f'_0) dx \right]$
for f_0, g_0 : winding number 0.

Both strategies lead to the same results.

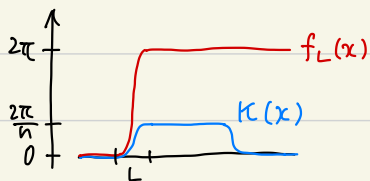
$$\gamma(f, g) = \exp 2\pi i \frac{k}{2} \left[\int_0^{2\pi} (f g' - g f') dx + f(0) \overset{\text{winding number}}{w_g} - w_f g(0) \right]$$

const. func. (nontriv. elem. of $\mathbb{Z}_2 \subset V(1)$ on entire space)

$$\begin{aligned} \text{e.g. } u(\tfrac{1}{2}) u(f) u(\tfrac{1}{2})^{-1} &= \gamma(\tfrac{1}{2}, f) u(f) \\ &= (-1)^{kw_f} u(f) \end{aligned}$$

$\leadsto (-1)^F = u(\tfrac{1}{2})$, $u(f)$ is fermionic if k and w_f odd.

$$\mathbb{Z}_n \subset \mathcal{V}(1)$$



$$P_a(0) = U(a\kappa) \circ U(a\kappa)^{-1}$$

$$0 \leq a < n$$

· fusion op.

$$U_{b,c}^L = U(f_L)^{\gamma(b,c)}$$

$$\gamma(b,c) := \begin{cases} 0 & (b+c < n) \\ 1 & (b+c \geq n) \end{cases}$$

carry
↑
if $\frac{a}{n} + \frac{b}{n} \geq 1$

{(★)}

$$\alpha(a,b,c) = \exp 2\pi i \left[\frac{k}{2} \frac{a}{n} \gamma(b,c) \right]$$

$k=2$
 $\leadsto \alpha(a,b,c)$ is a rep. of
 the generator of $H^3(\mathbb{Z}_n; \mathcal{V}(1)) \cong \mathbb{Z}_n$.

$$(-1)^F U_{b,c}^L (-1)^F = \gamma(b,c) U_{b,c}^L$$

↓

$$\gamma(b,c) = (-1)^{k\gamma(b,c)}$$

future directions :

- other abelian case ? $\mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2} \subset \mathcal{V}(1) \times \mathcal{V}(1)$ etc.
- non-ab. case ? $Co_0 \subset Spin(24)$ etc.